

Vinitha K

AI/ML Engineer

+919645315747 | vinithak885@gmail.com | India | Open to Hybrid | [LinkedIn](#)

SUMMARY

PROFILE SUMMARY AI Engineer with over 2 years of experience in designing, developing, and deploying ML/DL systems across healthcare and media analytics. Proven expertise in data preprocessing, model development, and production deployment using cloud-based solutions. Adept at leveraging LLMs, CNNs, and NLP techniques to solve complex business problems. Strong background in Python programming, data visualization, containerization (Docker), and cloud platforms (AWS)

WORK EXPERIENCE

AI ML Engineer, Tata Elxsi, Bangalore

May 2023 — Present

- Developed scalable AI/ML models for healthcare and media, translating research into production-ready tool. Heart Murmur Detection: Ensemble ML models using MFCC & wavelet transforms, improved diagnostic accuracy by 25%. Oral Cancer Screening: Implemented CNNs on RGB intraoral images with augmentation; supported early cancer detection.
- TV Content Tagging: Automated tagging using LLMs + audio fingerprinting, reduced manual effort by 80%.
- Integrated models into cloud platforms using Docker and CI/CD for scalable deployment.
- Collaborated cross-functionally to visualize insights and support informed decision-making.

EDUCATION

Cochin University of Science and Technology (CUSAT), Kochi -

Masters, M.Sc. Computer Science Specialization in Data science - (GPA: 9.2)

Aug 2021 — Jan 2023

Semester 1 - created a Indian rainfall visualisation of 100 year using matplotlib.

Semester 2 - Yolov3 based BMI calculator idea we have proposed.

Semester 3 - Audio assistant to visually disabled people using Yolov3

Semester 4 - Oral cancer detection using machine learning and deep learning techniques

Calicut University, Palakkad - Bachelors, Computer science

Jan 2021

During my bachelor I have done IoT based projects and web page designing using html , css, JavaScript.

GHSS OTTAPALAM, Ottapalam - Higher secondary , XIIth English

Jan 2018

I have started learning coding and identified my passion towards coding during these period.

LSN GHS OTTAPALAM , Ottapalam - High school, Xth Malayalam

Mar 2016

CERTIFICATIONS

[Computer vision master class](#), Udemy

Feb 2025

[Artificial intelligence A-Z 2025: Agentic AI, Gen AI and RL](#), Udemy

Jul 2025

[Complete MLOps Bootcamp With 10+ End To End ML Projects](#), Udemy

Dec 2024

[Statistical interpretation of medical research results](#), Udemy

Oct 2024

PROJECTS

Vehicle parked on Bus stop detection, [Link](#)

Aug 2025 — Present

- Implemented YOLOv7 model to accurately detect vehicles and static objects. Utilized sensor and GPS data to precisely determine the coordinates of the identified vehicles.

Synthetic dataset creation, Tata Elxsi , [Link](#), Bangalore

Jun 2025 — Aug 2025

- Generated cracked metal brackets using computer vision techniques and llm based approaches. Generated image to image generation using controlnet with stable diffusion model.

AI based content tagging fo INDIAN BROADCAST CHANNEL, [Link](#)

Dec 2024 — May 2025

- Developed an AI-powered pipeline to classify and tag broadcast content (commercials, programs, and promotions) for major Indian TV channels.
- Implemented audio and video-based content separation. Used audio fingerprinting and classification techniques to detect and segment advertisements.
- Leveraged Large Language Models (LLMs) for program and ad classification, and for structuring and standardizing news stories and sub-stories.
- Designed and deployed a scalable end-to-end production pipeline for automated content tagging and broadcast metadata generation.

- Cement packet detection using YOLOv7, [Link](#)

Oct 2024 — Dec 2024

 - Implemented Cement packet identification in the industry using YOLOv7 and image enhancement techniques. Achieved 98.2% accuracy through denoising methods.
- Person segmentation using RepViT

Oct 2024 — Dec 2024

 - Performed person segmentation in the cityscapes dataset using RepViT model.
- Car detection using YOLOv3 with MobileNet

Oct 2024 — Dec 2024

 - Developed a YOLOv3 with MobileNet model for car detection and classification.
- Heart Murmur Detection (Medical R&D)

May 2023 — Oct 2024

 - Designed and developed deep learning models from scratch and implemented transfer learning to detect heart murmurs from auscultation signals.
 - Extracted robust audio features using MFCCs, wavelet transforms, and other time-frequency domain techniques.
 - Applied a range of ML/DL algorithms and ensemble methods to enhance model accuracy and reliability.
 - Performed extensive data cleaning, augmentation, and preprocessing to ensure high-quality input for training and validation.
 - Conducted multiple research paper implementations and integrated findings into model improvement cycles.
 - Successfully completed two phases: R&D prototype and production-ready deployment.
- Oral cancer detection using ensembling method

Dec 2022 — May 2023

 - Implemented Convolutional Neural Networks (CNNs) to identify potential signs of oral cancer from RGB intraoral images, improving early detection capabilities.
 - Applied effective image preprocessing and augmentation techniques to enhance model generalization and robustness.
 - Contributed to the development of an AI-assisted screening tool.

SKILLS

Machine Learning	: Deep Learning, Machine Learning, Large Language Model, Natural Language Processing, Classification, Regression, Ensemble Learning
Frameworks & Libraries	: Pytorch, Tensorflow, Keras, Scikit-learn, OpenCV
Cloud & DevOps	: AWS, Docker, Ci/Cd, Google Cloud
Data Handling	: Numpy, Pandas, EHR, Data Pipelines, Flask API
Visualization & Tools	: Matplotlib, Seaborn, Streamlit, Jupyter Notebooks
Languages	: Hindi, Malayalam, Tamil, English